Detailed Solutions: MTL106/MTL2006 Tutorial-3

(Revised: Explicit Justifications, Mathematical Q9)

Q1. An airline estimates that 5% of the passengers who make reservations for a particular flight will not show up. To account for these no-shows, the airline sells 55 tickets for a flight that has only 53 seats. Assume that each passenger independently decides whether or not to show up. What is the probability that there will be a seat available for every passenger who arrives at the airport?

Solution: *Justification of Distribution:* A random variable X follows a Binomial distribution, denoted $X \sim \text{Bin}(n, p)$, if it represents the number of "successes" in n independent and identical Bernoulli trials, each having a success probability p. The probability mass function (PMF) is given by $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Here, there are $n = 55$ passengers (trials). Each passenger independently decides to show up (a "success") or not. The probability of showing up is constant at $p = 1 - 0.05 = 0.95$. Because these are independent trials with binary outcomes and a constant probability, the number of passengers who show up, X, exactly follows a Binomial distribution: $X \sim \text{Bin}(55, 0.95)$.

A seat is available for every passenger who arrives if $X \leq 53$.

$$\begin{aligned} P(X \leq 53) &= 1 - P(X = 54) - P(X = 55) \\ &= 1 - \binom{55}{54}(0.95)^{54}(0.05)^1 - \binom{55}{55}(0.95)^{55}(0.05)^0 \\ &= 1 - 55(0.95)^{54}(0.05) - (0.95)^{55} \approx 0.741 \end{aligned}$$

Q2. The probability that a single shot hits an aircraft is 0.05. Assume that the outcomes of different shots are independent, so that the number of hits in n shots follows a binomial distribution. How many shots must be fired so that the probability of obtaining at least two hits exceeds 0.95?

Solution: *Justification of Distribution:* As given, the outcomes are independent. Each shot is a trial with two possible outcomes (hit or miss) and a constant probability of hitting $p = 0.05$. Therefore, the total number of hits X in n independent shots satisfies the conditions for a Binomial distribution, $X \sim \text{Bin}(n, 0.05)$, with PMF $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.

We require $P(X \geq 2) > 0.95$, which implies $1 - P(X < 2) > 0.95 \implies P(X = 0) + P(X = 1) < 0.05$. Substituting the PMF formula:

$$\binom{n}{0}(0.05)^0(0.95)^n + \binom{n}{1}(0.05)^1(0.95)^{n-1} < 0.05$$

$$(0.95)^n + n(0.05)(0.95)^{n-1} < 0.05$$

Solving this inequality numerically for n yields $n \geq 94$. Thus, at least 94 shots must be fired.

Q3. For any $X \sim \mathcal{B}(n, p)$, show that $\mathbb{E}[\frac{1}{1+X}] = \frac{1-(1-p)^{n+1}}{(n+1)p}$

Solution: By definition, the PMF of a Binomial random variable $X \sim \mathcal{B}(n, p)$ is $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. By the fundamental definition of expected value for a discrete random variable,

$\mathbb{E}[g(X)] = \sum_{k=0}^{n} g(k)P(X=k)$. Thus:

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{1+X}\right] &= \sum_{k=0}^{n} \frac{1}{1+k}\binom{n}{k}p^k(1-p)^{n-k} \\
&= \sum_{k=0}^{n} \frac{1}{1+k}\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k} \\
&= \frac{1}{(n+1)p}\sum_{k=0}^{n} \frac{(n+1)!}{(k+1)!(n-k)!}p^{k+1}(1-p)^{n-k}
\end{aligned}$$

Let $j = k+1$. As k goes from 0 to n, j goes from 1 to $n+1$.

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{1+X}\right] &= \frac{1}{(n+1)p}\sum_{j=1}^{n+1}\binom{n+1}{j}p^j(1-p)^{(n+1)-j} \\
&= \frac{1}{(n+1)p}\left[\sum_{j=0}^{n+1}\binom{n+1}{j}p^j(1-p)^{(n+1)-j} - \binom{n+1}{0}p^0(1-p)^{n+1}\right] \\
&= \frac{1-(1-p)^{n+1}}{(n+1)p}
\end{aligned}$$

Q4. A communication system consists of n independent components. Each component functions correctly with probability p. The system is considered to operate effectively if at least half of its components are functioning. For what values of p is a 5-component system more likely to operate effectively than a 3-component system?

Solution: *Justification of Distribution:* Let X_n be the number of functioning components in an n-component system. Since the n components are independent, each has a binary state (functioning/failing), and each has a constant probability p of functioning, X_n rigorously follows a Binomial distribution: $X_n \sim \text{Bin}(n,p)$.

A 5-component system operates effectively if $X_5 \geq 3$. $P(X_5 \geq 3) = \binom{5}{3}p^3(1-p)^2 + \binom{5}{4}p^4(1-p) + \binom{5}{5}p^5 = 10p^3(1-p)^2 + 5p^4(1-p) + p^5$. A 3-component system operates effectively if $X_3 \geq 2$. $P(X_3 \geq 2) = \binom{3}{2}p^2(1-p) + \binom{3}{3}p^3 = 3p^2(1-p) + p^3$. We want $P(X_5 \geq 3) > P(X_3 \geq 2)$:

$$10p^3(1-p)^2 + 5p^4(1-p) + p^5 > 3p^2(1-p) + p^3$$

Since $p \in (0,1)$, we can divide by p^2:

$$10p(1-p)^2 + 5p^2(1-p) + p^3 > 3(1-p) + p$$

$$10p - 20p^2 + 10p^3 + 5p^2 - 5p^3 + p^3 > 3 - 2p$$

$$6p^3 - 15p^2 + 12p - 3 > 0 \implies 3(2p-1)(p-1)^2 > 0$$

Since $(p-1)^2 > 0$ for $p \neq 1$, we must have $2p - 1 > 0 \implies p > 1/2$. Thus, $p \in (0.5, 1)$.

Q5. Let $X \sim \mathcal{P}(\lambda)$ be a Poisson random variable with parameter λ. Prove the following: a) For every positive integer n $\mathbb{E}[X^n] = \lambda\mathbb{E}[(1+X)^{n-1}]$. b) Show that $P(X \text{ is even}) = \frac{1}{2}(1+e^{-2\lambda})$ and $\mathbb{P}(X \text{ is odd}) = \frac{1}{2}(1-e^{-2\lambda})$.

Solution: The PMF of a Poisson random variable is $P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$. a) $\mathbb{E}[X^n] = \sum_{k=0}^{\infty} k^n \frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=1}^{\infty} k^n \frac{e^{-\lambda}\lambda^k}{k!} = \lambda\sum_{k=1}^{\infty} k^{n-1}\frac{e^{-\lambda}\lambda^{k-1}}{(k-1)!}$. Let $j = k-1$. $\mathbb{E}[X^n] = \lambda\sum_{j=0}^{\infty}(j+1)^{n-1}\frac{e^{-\lambda}\lambda^j}{j!} = \lambda\mathbb{E}[(X+1)^{n-1}]$.

b) $P(X \text{ is even}) = \sum_{k=0}^{\infty} P(X = 2k) = \sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^{2k}}{(2k)!}$. We know the Maclaurin series expansions: $e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$ and $e^{-\lambda} = \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!}$. Summing them gives $e^{\lambda} + e^{-\lambda} = 2\sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!}$. Thus, $P(X \text{ is even}) = e^{-\lambda}\sum_{k=0}^{\infty} \frac{\lambda^{2k}}{(2k)!} = e^{-\lambda}\left(\frac{e^{\lambda}+e^{-\lambda}}{2}\right) = \frac{1+e^{-2\lambda}}{2}$. Similarly, $P(X \text{ is odd}) = 1 - P(X \text{ is even}) = \frac{1-e^{-2\lambda}}{2}$.

Q6. Suppose that the average number of accidents occurring each week on a particular highway is 5. Assuming that the number of accidents in a week follows a Poisson distribution, calculate the probability that at least two accidents occur during a given week.
Solution: *Justification of Distribution:* A Poisson distribution models the number of events occurring in a fixed interval of time or space, provided these events happen with a known constant mean rate and independently of the time since the last event. The PMF is $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$. The problem specifies that accidents occur with an average rate of $\lambda = 5$ per week. Assuming independence, the number of accidents X in a week satisfies $X \sim \text{Poisson}(5)$.

We want $P(X \geq 2)$. $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - \frac{e^{-5}5^0}{0!} - \frac{e^{-5}5^1}{1!} = 1 - e^{-5}(1 + 5) = 1 - 6e^{-5} \approx 0.9596$.

Q7. A manufacturer produces light bulbs that are packaged in boxes of 1000. Quality control studies indicate that 0.5% of the light bulbs produced are defective. Determine the percentage of boxes that contain: (a) no defective light bulbs, (b) two or more defective light bulbs.
Solution: *Justification of Distribution:* Let Y be the number of defective bulbs in a box of $n = 1000$. Since each bulb is independently defective with probability $p = 0.005$, the exact distribution is $Y \sim \text{Bin}(1000, 0.005)$. For a Binomial distribution where the number of trials n is very large and the probability of success p is very small, the probabilities are mathematically well-approximated by a Poisson distribution with the rate parameter $\lambda = np$. Here, $\lambda = 1000 \times 0.005 = 5$. Thus, we model the number of defective bulbs X as $X \sim \text{Poisson}(5)$.

(a) $P(X = 0) = \frac{e^{-5}5^0}{0!} = e^{-5} \approx 0.0067$ (or 0.67%). (b) $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-5} - \frac{e^{-5}5^1}{1!} = 1 - 6e^{-5} \approx 0.9596$ (or 95.96%).

Q8. A large batch of automobile tires contains 5% defective tires. Six tires are selected one at a time to equip a car. The selection process continues until 6 good tires have been found. i) Find the probability that 3 defective tires are found before the 6th good tire. ii) Find the mean and variance of the number of defective tires found before the 6th good tire.
Solution: *Justification of Distribution:* A Negative Binomial random variable $X \sim \text{NB}(r, p)$ counts the number of failures before the r-th success in a sequence of independent Bernoulli trials with a constant success probability p. The PMF is given by $P(X = k) = \binom{k+r-1}{k} p^r (1-p)^k$. In this scenario, a "success" is finding a good tire ($p = 0.95$), and a "failure" is finding a defective tire ($1 - p = 0.05$). We terminate the process upon achieving exactly $r = 6$ successes. Therefore, the number of defective tires found (failures), X, perfectly maps to a Negative Binomial distribution: $X \sim \text{NB}(r = 6, p = 0.95)$.

i) $P(X = 3) = \binom{3+6-1}{3}(0.95)^6(0.05)^3 = \binom{8}{3}(0.95)^6(0.05)^3 = 56 \times 0.73509 \times 0.000125 \approx 0.0051$. ii) The mathematical expectation for a Negative Binomial distribution is $\mathbb{E}[X] = \frac{r(1-p)}{p} = \frac{6 \times 0.05}{0.95} = \frac{6}{19} \approx 0.316$. The variance is $\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{6 \times 0.05}{0.95^2} = \frac{120}{361} \approx 0.332$.

Q9. Let $X \sim \mathcal{B}_N(r, p)$ denote a negative binomial random variable representing the number of failures before the r-th success, and let $Y \sim \mathcal{B}(n, p)$ denote a binomial random variable with parameters n and p. Show that $P(X > n) = P(Y < r)$.
Solution: *Note on Notation:* For the equality $P(X > n) = P(Y < r)$ to hold algebraically given $Y \sim \text{Bin}(n, p)$, the random variable X must mathematically represent the *total number of trials* required to achieve the r-th success (rather than just the number of failures). If X

were strictly the number of failures, Y would need to be $\text{Bin}(n+r,p)$. Assuming the standard algebraic form intended by the problem's equality, let X denote the total trials.

Let X have the PMF $P(X=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r}$ for $k\ge r$. Let Y have the PMF $P(Y=j)=\binom{n}{j}p^j(1-p)^{n-j}$ for $0\le j\le n$.

We must prove that $1-P(X\le n)=P(Y<r)$, which translates to:

$$1-\sum_{k=r}^{n}\binom{k-1}{r-1}p^r(1-p)^{k-r}=\sum_{j=0}^{r-1}\binom{n}{j}p^j(1-p)^{n-j}$$

Proof by Mathematical Induction on n:

Base Case ($n=r$): Left-Hand Side (LHS): $1-\binom{r-1}{r-1}p^r(1-p)^0=1-p^r$. Right-Hand Side (RHS): $\sum_{j=0}^{r-1}\binom{r}{j}p^j(1-p)^{r-j}$. Since $\sum_{j=0}^{r}\binom{r}{j}p^j(1-p)^{r-j}=(p+(1-p))^r=1^r=1$, the sum up to $r-1$ is exactly 1 minus the final $j=r$ term: $1-\binom{r}{r}p^r(1-p)^0=1-p^r$. LHS = RHS, so the base case holds.

Inductive Step: Assume the equality holds for an arbitrary integer $m\ge r$:

$$1-\sum_{k=r}^{m}\binom{k-1}{r-1}p^r(1-p)^{k-r}=\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j}$$

We must prove it holds for $m+1$. We start by expressing the LHS for $m+1$ and isolating the final term:

$$\begin{aligned}\text{LHS}_{m+1}&=1-\sum_{k=r}^{m+1}\binom{k-1}{r-1}p^r(1-p)^{k-r}\\&=\left(1-\sum_{k=r}^{m}\binom{k-1}{r-1}p^r(1-p)^{k-r}\right)-\binom{m}{r-1}p^r(1-p)^{m+1-r}\end{aligned}$$

Substitute the bracketed expression using our inductive hypothesis:

$$\text{LHS}_{m+1}=\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j}-\binom{m}{r-1}p^r(1-p)^{m+1-r}$$

Now, we independently expand the RHS for $m+1$:

$$\text{RHS}_{m+1}=\sum_{j=0}^{r-1}\binom{m+1}{j}p^j(1-p)^{m+1-j}$$

Using the combinatorial identity $\binom{m+1}{j}=\binom{m}{j}+\binom{m}{j-1}$ for $j\ge 1$:

$$\text{RHS}_{m+1}=\binom{m+1}{0}p^0(1-p)^{m+1}+\sum_{j=1}^{r-1}\left[\binom{m}{j}+\binom{m}{j-1}\right]p^j(1-p)^{m+1-j}$$

Distribute and absorb the $j=0$ term back into the first sum:

$$=\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m+1-j}+\sum_{j=1}^{r-1}\binom{m}{j-1}p^j(1-p)^{m+1-j}$$

Factor out $(1-p)$ from the first sum and p from the second sum:

$$=(1-p)\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j}+p\sum_{j=1}^{r-1}\binom{m}{j-1}p^{j-1}(1-p)^{m-(j-1)}$$

Shift the index of the second sum by letting $i = j - 1$:

$$= (1-p)\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j} + p\sum_{i=0}^{r-2}\binom{m}{i}p^i(1-p)^{m-i}$$

Add and subtract the missing $i = r - 1$ term inside the bracket to match the limits:

$$= (1-p)\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j} + p\left[\sum_{i=0}^{r-1}\binom{m}{i}p^i(1-p)^{m-i} - \binom{m}{r-1}p^{r-1}(1-p)^{m-r+1}\right]$$

Combine the two matching summation terms:

$$= \big((1-p)+p\big)\sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j} - p\binom{m}{r-1}p^{r-1}(1-p)^{m+1-r}$$

Because $(1-p)+p = 1$, this simplifies exactly to:

$$= \sum_{j=0}^{r-1}\binom{m}{j}p^j(1-p)^{m-j} - \binom{m}{r-1}p^r(1-p)^{m+1-r}$$

Since $\text{LHS}_{m+1} = \text{RHS}_{m+1}$, the algebraic equality is rigorously proven for all $n \geq r$ by mathematical induction.

Q10. A stick of length 1 is broken at a point U, where U is uniformly distributed over the interval (0, 1). Let L denote the length of the piece that contains the point $\frac{1}{3}$. Determine $\mathbb{E}(L)$ and $Var(L)$.
Solution: The stick is broken at $U \sim U(0,1)$, meaning the probability density function (PDF) is $f(u) = 1$ for $u \in (0,1)$. The piece containing $1/3$ is $[0, U]$ if $U > 1/3$, and $[U, 1]$ if $U \leq 1/3$. So $L(U) = U$ if $U \in (1/3, 1]$, and $L(U) = 1 - U$ if $U \in [0, 1/3]$.

$$\begin{aligned}
\mathbb{E}[L] &= \int_0^1 L(u)f(u)du = \int_0^{1/3}(1-u)(1)du + \int_{1/3}^1 u(1)du \\
&= \left[u - \frac{u^2}{2}\right]_0^{1/3} + \left[\frac{u^2}{2}\right]_{1/3}^1 \\
&= \left(\frac{1}{3} - \frac{1}{18}\right) + \left(\frac{1}{2} - \frac{1}{18}\right) = \frac{5}{18} + \frac{8}{18} = \frac{13}{18}.
\end{aligned}$$

For variance, we find $\mathbb{E}[L^2]$:

$$\begin{aligned}
\mathbb{E}[L^2] &= \int_0^{1/3}(1-u)^2du + \int_{1/3}^1 u^2du = \left[-\frac{(1-u)^3}{3}\right]_0^{1/3} + \left[\frac{u^3}{3}\right]_{1/3}^1 \\
&= -\frac{(2/3)^3}{3} + \frac{1}{3} + \frac{1}{3} - \frac{(1/3)^3}{3} = \frac{5}{9}.
\end{aligned}$$

$\text{Var}(L) = \mathbb{E}[L^2] - (\mathbb{E}[L])^2 = \frac{5}{9} - \left(\frac{13}{18}\right)^2 = \frac{20}{36} - \frac{169}{324} = \frac{11}{324}$.

Q11. A point is selected at random from the interval [-1,3], with all points being equally likely. If X denotes the selected point and $Y = X^2$ determine the probability density function of Y.
Solution: $X \sim U[-1, 3]$. The PDF of a uniform distribution over $[a, b]$ is $f_X(x) = \frac{1}{b-a}$. Thus, $f_X(x) = 1/4$ for $x \in [-1, 3]$. $Y = X^2$. The range of Y is $[0, 9]$. We find the cumulative

distribution function (CDF) $F_Y(y) = P(Y \le y)$. For $y \in (0,1)$, $P(Y \le y) = P(X^2 \le y) = P(-\sqrt{y} \le X \le \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{4}dx = \frac{2\sqrt{y}}{4} = \frac{\sqrt{y}}{2}$. For $y \in (1,9)$, $P(Y \le y) = P(-1 \le X \le \sqrt{y}) = \int_{-1}^{\sqrt{y}} \frac{1}{4}dx = \frac{\sqrt{y}+1}{4}$. Differentiating the CDF to obtain the PDF $f_Y(y) = \frac{d}{dy}F_Y(y)$: If $0 < y \le 1$: $f_Y(y) = \frac{d}{dy}\left(\frac{\sqrt{y}}{2}\right) = \frac{1}{4\sqrt{y}}$. If $1 < y < 9$: $f_Y(y) = \frac{d}{dy}\left(\frac{\sqrt{y}+1}{4}\right) = \frac{1}{8\sqrt{y}}$. $f_Y(y) = 0$ otherwise.

Q12. Suppose X and Y are random variables, where X denotes the speed and $Y = \frac{180}{X}$ denotes the travel time (duration). If X is uniformly distributed on the interval [30, 60], find the probability density function (pdf) of Y.
Solution: $X \sim U[30,60]$. The PDF is $f_X(x) = \frac{1}{60-30} = 1/30$ for $x \in [30,60]$. We have the transformation $Y = 180/X$. The range of Y maps from $[30,60]$ to $[180/60, 180/30] = [3,6]$. By the transformation method for PDFs, $f_Y(y) = f_X(x)\left|\frac{dX}{dY}\right|$. $X = 180/Y \implies \frac{dX}{dY} = -\frac{180}{Y^2}$. $f_Y(y) = \frac{1}{30}\left|-\frac{180}{y^2}\right| = \frac{1}{30} \times \frac{180}{y^2} = \frac{6}{y^2}$ for $3 \le y \le 6$, and 0 otherwise.

Q13. Consider the marks obtained by students in the Mathematics examination. Assume that the marks are normally distributed with a mean of 56 and a standard deviation of 18. It is observed that 10% of the students receive an A grade, while 5% of the students fail the course. i) Determine the minimum mark required to obtain an A grade. ii) Determine the minimum mark required to pass the course.
Solution: *Justification of Distribution:* The problem explicitly states that marks X follow a Normal distribution, parameterized by its mean $\mu = 56$ and standard deviation $\sigma = 18$, thus $X \sim N(56, 18^2)$. The PDF of a Normal distribution is $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$. We standardize to the standard normal $Z \sim N(0,1)$ using $Z = \frac{X-\mu}{\sigma}$.

i) Minimum mark for A grade (top 10%): We need x_A such that $P(X \ge x_A) = 0.10 \implies P\left(Z \ge \frac{x_A-56}{18}\right) = 0.10$. From standard normal tables, the Z-score for the upper 10th percentile is $Z_{0.10} \approx 1.28$. $x_A = 56 + 1.28 \times 18 = 56 + 23.04 = 79.04$. ii) Minimum mark to pass (bottom 5% fail): We need x_P such that $P(X < x_P) = 0.05 \implies P\left(Z < \frac{x_P-56}{18}\right) = 0.05$. From standard normal tables, $Z_{0.05} \approx -1.645$. $x_P = 56 - 1.645 \times 18 = 56 - 29.61 = 26.39$.

Q14. A manufacturer produces bolts whose diameters are normally distributed with a mean of 1.2 inches and a standard deviation of 0.01 inches. To satisfy quality specifications, each bolt must have a diameter between 1.19 inches and 1.21 inches. What percentage of the bolts produced do not meet these specifications?
Solution: Let X be the diameter of the bolts. By the problem description, $X \sim N(\mu = 1.2, \sigma^2 = 0.01^2)$. Specifications are satisfied when $1.19 \le X \le 1.21$. First, calculate the probability of a bolt meeting the specifications using standard normal variables $Z = \frac{X-\mu}{\sigma}$: $P(1.19 \le X \le 1.21) = P\left(\frac{1.19-1.2}{0.01} \le Z \le \frac{1.21-1.2}{0.01}\right) = P(-1 \le Z \le 1)$. Using standard normal distribution tables, $P(-1 \le Z \le 1) \approx 0.6827$. The percentage of bolts not meeting specifications is $1 - 0.6827 = 0.3173 = 31.73\%$.

Q15. Let $f(x)$ be the probability density function of a normally distributed random variable with mean μ and variance σ^2 Show that the points $x = \mu - \sigma$ and $x = \mu + \sigma$ are points of inflection of the graph of $f(x)$. In other words, prove that $f''(x) = 0$ when $x = \mu - \sigma$ and $x = \mu + \sigma$.
Solution: The exact mathematical definition of the Normal PDF is $f(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. Taking the first derivative using the chain rule: $f'(x) = f(x) \cdot \left(-\frac{2(x-\mu)}{2\sigma^2}\right) = f(x) \cdot \left(-\frac{x-\mu}{\sigma^2}\right)$. Applying the product rule for the second derivative: $f''(x) = f'(x)\left(-\frac{x-\mu}{\sigma^2}\right) + f(x)\left(-\frac{1}{\sigma^2}\right) = f(x)\left[\frac{(x-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\right] = \frac{f(x)}{\sigma^2}\left[\left(\frac{x-\mu}{\sigma}\right)^2 - 1\right]$. Setting $f''(x) = 0$, since $f(x)$ is strictly positive, we get $\left(\frac{x-\mu}{\sigma}\right)^2 = 1 \implies \frac{x-\mu}{\sigma} = \pm 1 \implies x = \mu \pm \sigma$. Thus, $x = \mu - \sigma$ and $x = \mu + \sigma$ are precisely the points of inflection.

Q16. Let X be a normally distributed random variable with mean μ and variance σ^2. If $Y = e^X$, determine the probability density function of Y.

Solution: $X \sim N(\mu, \sigma^2)$. The variable $Y = e^X$ follows a log-normal distribution. To derive its PDF $f_Y(y)$, we first establish its CDF. Since $e^x > 0$, Y takes values in $(0, \infty)$. For $y > 0$, $F_Y(y) = P(Y \leq y) = P(e^X \leq y) = P(X \leq \ln y) = F_X(\ln y)$, where F_X is the CDF of the Normal distribution. Differentiating with respect to y gives the PDF via the chain rule: $f_Y(y) = \frac{d}{dy} F_X(\ln y) = f_X(\ln y) \cdot \frac{d}{dy}(\ln y) = \frac{1}{y} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right)$. For $y \leq 0$, the probability is zero, so $f_Y(y) = 0$.

Q17. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 1$. i) Find the probability that the repair time exceeds 2 hours. ii) Given that the repair has already lasted more than 2 hours, find the probability that it requires at least 3 hours to complete.

Solution: *Distribution Definition:* A random variable T follows an Exponential distribution with rate parameter $\lambda > 0$ if its PDF is given by $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$, representing the continuous waiting time until an event occurs. The CDF is $P(T \leq t) = 1 - e^{-\lambda t}$, giving a survival function $P(T > t) = e^{-\lambda t}$. Here, $T \sim \text{Exp}(1)$, so $P(T > t) = e^{-t}$. i) The probability that repair time exceeds 2 hours is $P(T > 2) = \int_2^\infty e^{-t} dt = e^{-2} \approx 0.1353$. ii) Using the definition of conditional probability: $P(T > 3 \mid T > 2) = \frac{P(T>3 \text{ and } T>2)}{P(T>2)} = \frac{P(T>3)}{P(T>2)} = \frac{e^{-3}}{e^{-2}} = e^{-1} \approx 0.3679$. This demonstrates that the probability of lasting one more hour is completely independent of the two hours that have already passed.

Q18. The lifetime (in years) of a radio is exponentially distributed with a mean of 8 years. If Jones purchases a used radio, what is the probability that it will continue to function for at least another 8 years?

Solution: As defined in Q17, for $T \sim \text{Exp}(\lambda)$, the expected value mathematically is $\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t} dt = 1/\lambda$. Given the mean is 8, we have $1/\lambda = 8 \implies \lambda = 1/8$. Using conditional probability, the probability that the radio functions for at least another 8 years given it has survived until now (t years) is computed as follows: $P(T > t + 8 \mid T > t) = \frac{P(T>t+8)}{P(T>t)} = \frac{e^{-\lambda(t+8)}}{e^{-\lambda t}} = e^{-\lambda \times 8} = e^{-(1/8)\times 8} = e^{-1} \approx 0.3679$. This result depends only on the required 8 additional years, irrespective of the starting time t.

Q19. The time to failure T (in operating hours) of a compressor is a continuous random variable with probability density function $f(t) = \frac{0.002}{(0.001t+1)^3}, t \geq 0$ and 0 otherwise. Determine the following: i) Find the failure rate (hazard) function of the compressor. ii) Determine the reliability of the compressor for an operating life of 100 hours. iii) Determine the design life of the compressor if a reliability of 0.95 is required. iv) Determine the mean time to failure (MTTF) of the compressor.

Solution: The fundamental relationship between PDF $f(t)$ and reliability function $R(t)$ is $R(t) = P(T > t) = \int_t^\infty f(u) du$. Here, $R(t) = \int_t^\infty 0.002(0.001u + 1)^{-3} du$. Let $w = 0.001u + 1 \implies dw = 0.001du$. $R(t) = \int_{0.001t+1}^\infty 2w^{-3} dw = \left[-w^{-2}\right]_{0.001t+1}^\infty = (0.001t + 1)^{-2}$. i) Hazard function definition is $h(t) = \frac{f(t)}{R(t)} = \frac{0.002(0.001t+1)^{-3}}{(0.001t+1)^{-2}} = \frac{0.002}{0.001t+1}$. ii) Reliability for 100 hours: $R(100) = (0.001 \times 100 + 1)^{-2} = (1.1)^{-2} \approx 0.8264$. iii) Design life for reliability 0.95 requires solving $R(t) = 0.95 \implies (0.001t + 1)^{-2} = 0.95 \implies 0.001t + 1 = (0.95)^{-0.5} \approx 1.02598 \implies t \approx 25.98$ hours. iv) By definition, $\text{MTTF} = \mathbb{E}[T] = \int_0^\infty R(t) dt$. $\text{MTTF} = \int_0^\infty (0.001t + 1)^{-2} dt = \left[-\frac{(0.001t+1)^{-1}}{0.001}\right]_0^\infty = 1000$ hours.

Q20. Let X be a uniformly distributed random variable on the interval (0,2). Determine the failure rate (hazard) function of X. If $Y = 3X$, find the failure rate (hazard) function of Y.

Solution: $X \sim U(0, 2)$. Its PDF is $f_X(x) = 1/2$ and CDF is $F_X(x) = x/2$ for $x \in (0, 2)$.

Reliability function $R_X(x) = 1 - F_X(x) = 1 - x/2 = \frac{2-x}{2}$. Failure rate of X: $h_X(x) = \frac{f_X(x)}{R_X(x)} = \frac{1/2}{(2-x)/2} = \frac{1}{2-x}$ for $0 < x < 2$. For $Y = 3X$, the transformation scales the support linearly to $Y \sim U(0, 6)$. The PDF is $f_Y(y) = 1/6$, CDF is $F_Y(y) = y/6$, and Reliability is $R_Y(y) = \frac{6-y}{6}$. Failure rate of Y: $h_Y(y) = \frac{f_Y(y)}{R_Y(y)} = \frac{1/6}{(6-y)/6} = \frac{1}{6-y}$ for $0 < y < 6$.

Q21. The time (in hours) taken by a customer to choose a car at a Honda showroom is gamma distributed with a mean of 2 hours and a variance of 2 hours^2. i) What is the probability that a customer takes no more than 4.5 hours to choose a car? ii) Find the 90th percentile of the time required to choose a car.
Solution: *Distribution Definition:* A Gamma random variable $X \sim \text{Gamma}(\alpha, \lambda)$ is defined for $x > 0$ with PDF $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$. The parameters are shape α and rate λ. Its mathematical mean is $\mathbb{E}[X] = \alpha/\lambda$ and variance is $\text{Var}(X) = \alpha/\lambda^2$. Given $\alpha/\lambda = 2$ and $\alpha/\lambda^2 = 2$. Dividing the mean by the variance yields $\lambda = 1$. Substituting λ back gives $\alpha = 2$. With these parameters, $\Gamma(2) = 1! = 1$, so the PDF simplifies to $f_X(x) = xe^{-x}$ for $x > 0$. i) The probability is the CDF evaluated at 4.5: $P(X \leq 4.5) = \int_0^{4.5} xe^{-x} dx$. Using integration by parts: $\int_0^{4.5} xe^{-x} dx = [-xe^{-x} - e^{-x}]_0^{4.5} = (-4.5e^{-4.5} - e^{-4.5}) - (-0 - 1) = 1 - 5.5e^{-4.5} \approx 0.9389$. ii) The 90th percentile x_{90} solves $P(X \leq x_{90}) = 0.90 \implies \int_0^{x_{90}} xe^{-x} dx = 0.90$. $1 - (x_{90} + 1)e^{-x_{90}} = 0.90 \implies (x_{90} + 1)e^{-x_{90}} = 0.10$. Solving this nonlinear equation numerically yields $x_{90} \approx 3.89$ hours.

Q22. Prove that the reciprocal of a standard Cauchy random variable is also a standard Cauchy random variable.
Solution: By definition, a standard Cauchy random variable X has the PDF $f_X(x) = \frac{1}{\pi(1+x^2)}$ over the domain $x \in \mathbb{R}$. Let us define the transformation $Y = 1/X$. Since X spans the real line (excluding an event of measure zero at $X = 0$), Y also spans the real line. The inverse transformation is $x = 1/y$. The absolute value of the Jacobian determinant is $|dx/dy| = |-1/y^2| = 1/y^2$. Using the PDF transformation formula $f_Y(y) = f_X(x(y))|dx/dy|$, we get: $f_Y(y) = \frac{1}{\pi(1+(1/y)^2)} \cdot \frac{1}{y^2} = \frac{1}{\pi\left(\frac{y^2+1}{y^2}\right)y^2} = \frac{1}{\pi(y^2+1)}$. Because $f_Y(y)$ is identical in mathematical form to $f_X(x)$, the random variable Y follows the exact same standard Cauchy distribution.

Q23. Let X be a random variable having probability density function $f(x) = kx^{-\frac{1}{2}}(1-x)^{\frac{1}{2}}; 0 < x < 1$. where k is a normalizing constant. Determine the value of k. Find the mean and variance of X.
Solution: *Distribution Definition:* A continuous random variable follows a Beta distribution over $(0, 1)$ with shape parameters $\alpha, \beta > 0$ if its PDF is $f(x) = \frac{1}{B(\alpha,\beta)} x^{\alpha-1}(1-x)^{\beta-1}$, where the Beta function $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$. Comparing the given $f(x) \propto x^{-1/2}(1-x)^{1/2}$ to the standard form: $\alpha - 1 = -1/2 \implies \alpha = 1/2$. $\beta - 1 = 1/2 \implies \beta = 3/2$. The normalizing constant k must be $1/B(1/2, 3/2)$ to ensure the integral over $(0, 1)$ equals 1. $k = \frac{\Gamma(1/2+3/2)}{\Gamma(1/2)\Gamma(3/2)} = \frac{\Gamma(2)}{\Gamma(1/2)\Gamma(3/2)}$. Using the properties $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(x+1) = x\Gamma(x) \implies \Gamma(3/2) = \frac{1}{2}\sqrt{\pi}$: $k = \frac{1!}{\sqrt{\pi}\cdot\frac{1}{2}\sqrt{\pi}} = \frac{1}{\pi/2} = \frac{2}{\pi}$. For a Beta distribution, the mathematically derived mean and variance are $\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}$ and $\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. Mean $\mathbb{E}[X] = \frac{1/2}{1/2+3/2} = \frac{1/2}{2} = \frac{1}{4}$. Variance $\text{Var}(X) = \frac{(1/2)(3/2)}{(2)^2(2+1)} = \frac{3/4}{4\times3} = \frac{3/4}{12} = \frac{1}{16}$.

Q24. Determine whether the continuous uniform distribution is a special case of the beta distribution. If so, identify the corresponding values of the parameters.
Solution: The beta distribution PDF mathematically is $f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$ defined for $0 < x < 1$. The standard continuous uniform distribution on the interval $(0, 1)$ is characterized by a constant PDF $f(x) = 1$ for $0 < x < 1$. To see if the Beta distribution reduces to the Uniform,

we attempt to find $\alpha, \beta > 0$ such that the x dependencies vanish. This requires the exponents to be zero: $\alpha - 1 = 0 \implies \alpha = 1$ $\beta - 1 = 0 \implies \beta = 1$ Substituting these parameters into the Beta PDF: $f(x) = \frac{x^{1-1}(1-x)^{1-1}}{B(1,1)} = \frac{x^0(1-x)^0}{\Gamma(1)\Gamma(1)/\Gamma(2)} = \frac{1}{(1 \cdot 1)/1} = 1$. Because evaluating the Beta PDF at $\alpha = 1, \beta = 1$ yields exactly the uniform PDF $f(x) = 1$, the continuous uniform distribution is mathematically proven to be a special case of the beta distribution.